Week 8

Database Relationship Analysis using Joins

Scenario:

The e-commerce company maintains customer, product, order, and payment details in different tables. The management team needs to combine these tables to analyse customer purchases, order information, and payment status. Students will use SQL Joins to fetch useful details from multiple tables.

1. Combine Related Tables Using Joins

Customer Table

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID CUSTOMER_NAME       EMAIL                        PHONE ADDRESS                                            REGISTRAT
----------- ------------------- ---------------------------- ---------- ---------------------------------------------- ---------
        101 Arun Kumar Raj      arun@gmail.com               9876532101 Chennai                                        10-JAN-25
        102 Priya Sharma        priya@gmail.com              9876532102 Chennai                                        15-FEB-25
        103 Rahul Raj           rahul@gmail.com              9876532403 Chennai                                        20-MAR-25
        104 Divya Devi          divya@gmail.com              9876532104 Chennai                                        05-APR-25
```

Orders Table

```
SQL> SELECT * FROM Orders;

ORDER_ID             CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- ----------- --------- ------------ --------------------
111                          101 10-JAN-25         2500 Delivered
112                          102 15-FEB-25          307 Shipped
113                          103 20-MAR-25         4800 Pending
114                          104 05-APR-25          400 Pending
115                          105 12-MAY-25         5071 Delivered
```

Product Table

```
SQL> SELECT * FROM Product;

PRODUCT PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
------- -------------------- ------- ---------- --------------
P001    Ball Pen             101             20              5
P002    Gel Pen              102             15             10
P004    Pencil Box           104             60             20
P005    Spiral Notebook      105             60             20
P016    Notebook             102             55             40
P003    Pencil               101              8            120

6 rows selected.
```

Payment Table

```
SQL> SELECT * FROM Payment;

PAYMENT_ID ORDER_ID             PAYMENT_D PAYMENT_MODE        PAYMENT_STATUS  TRANSACTION_AMOUNT
---------- -------------------- --------- ------------------- --------------- ------------------
P101       111                  11-JUL-26 UPI                 Success                       2500
P102       112                  12-JUL-26 Card                Success                       1800
P103       113                  13-JUL-26 Net Banking         Pending                       3200
P104       114                  14-JUL-26 Cash                Success                       1500
P105       115                  15-JUL-26 UPI                 Success                       2200
```

2. Implement INNER JOIN

1. Display customer details along with their orders.

```
SQL> SELECT c.customer_id, c.customer_name, c.email, o.order_id, o.order_date, o.total_amount, o.order_status FROM customer c JOIN orders o ON c.customer_id = o.custo
mer_id;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------------------
EMAIL                          ORDER_ID             ORDER_DAT TOTAL_AMOUNT
------------------------------ -------------------- --------- ------------
ORDER_STATUS
--------------------
        101 Arun Kumar Raj
arun@gmail.com                 111                  10-JAN-25         2500
Delivered

        102 Priya Sharma
priya@gmail.com                112                  15-FEB-25          307
Shipped

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------------------
EMAIL                          ORDER_ID             ORDER_DAT TOTAL_AMOUNT
------------------------------ -------------------- --------- ------------
ORDER_STATUS
--------------------

        103 Rahul Raj
rahul@gmail.com                113                  20-MAR-25         4800
Pending

        104 Divya Devi
divya@gmail.com                114                  05-APR-25          400

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------------------
EMAIL                          ORDER_ID             ORDER_DAT TOTAL_AMOUNT
------------------------------ -------------------- --------- ------------
ORDER_STATUS
--------------------
Pending
```

2. Display order details with payment information.

```
SQL> SELECT o.order_id, o.order_date, o.total_amount, o.order_status, p.payment_id, p.payment_mode, p.payment_status, p.transaction_amount FROM orders o JOIN payment
p ON o.order_id = p.order_id;

ORDER_ID             ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
-------------------- --------- ------------ -------------------- ----------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------
111                  10-JAN-25         2500 Delivered            P101
UPI                  Success             2500

112                  15-FEB-25          307 Shipped              P102
Card                 Success             1800

113                  20-MAR-25         4800 Pending              P103
Net Banking          Pending             3200


ORDER_ID             ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
-------------------- --------- ------------ -------------------- ----------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------
114                  05-APR-25          400 Pending              P104
Cash                 Success             1500

115                  12-MAY-25         5071 Delivered            P105
UPI                  Success             2200
```

3. Retrieve products purchased by customers.

```
SQL> SELECT c.customer_name, o.order_id, p.product_name, od.quantity, od.price FROM customer c JOIN orders o ON c.customer_id = o.customer_id JOIN order_details od ON
 o.order_id = od.order_id JOIN product p ON od.product_id = p.product_id;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- --------------------
PRODUCT_NAME
------------------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Arun Kumar Raj                                     111
Ball Pen
         5       1000

Priya Sharma                                       112
Gel Pen
        10       2000

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- --------------------
PRODUCT_NAME
------------------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Divya Devi                                         114
Pencil Box
        20       4000

Rahul Raj                                          113
Pencil

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- --------------------
PRODUCT_NAME
------------------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
        15       3000
```

3. Implement LEFT JOIN

1. Display all customers including customers who have not placed orders.

```
SQL> SELECT C.*, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                          PHONE
------------------------------ ----------
ADDRESS                                            REGISTRAT
-------------------------------------------------- ---------
ORDER_ID    ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- --------- ------------ --------------------
        101 Arun Kumar Raj
arun@gmail.com                 9876532101
Chennai                                            10-JAN-25
111         10-JAN-25         2500 Delivered


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                          PHONE
------------------------------ ----------
ADDRESS                                            REGISTRAT
-------------------------------------------------- ---------
ORDER_ID    ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- --------- ------------ --------------------
        102 Priya Sharma
priya@gmail.com                9876532102
Chennai                                            15-FEB-25
112         15-FEB-25          307 Shipped


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                          PHONE
------------------------------ ----------
ADDRESS                                            REGISTRAT
-------------------------------------------------- ---------
ORDER_ID    ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- --------- ------------ --------------------
        103 Rahul Raj
rahul@gmail.com                9876532403
Chennai                                            20-MAR-25
113         20-MAR-25         4800 Pending


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                          PHONE
------------------------------ ----------
ADDRESS                                            REGISTRAT
-------------------------------------------------- ---------
ORDER_ID    ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- --------- ------------ --------------------
        104 Divya Devi
divya@gmail.com                9876532104
```

2. Find customers without any purchases.

```
SQL> SELECT C.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.ORDER_ID IS NULL;

no rows selected
```

3. Display all products including products with no sales.

```
SQL> SELECT P.*, OD.ORDER_ID, OD.QUANTITY FROM PRODUCT P LEFT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY ORDER_ID              QUANTITY
------ ---------- -------------- -------------------- ----------
P001
Ball Pen
101             20              5 111                           5

P002
Gel Pen
102             15             10 112                          10

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY ORDER_ID              QUANTITY
------ ---------- -------------- -------------------- ----------

P003
Pencil
101              8            120 113                          15

P004
Pencil Box

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGOR      PRICE STOCK_QUANTITY ORDER_ID              QUANTITY
------ ---------- -------------- -------------------- ----------
104             60             20 114                          20

P005
Spiral Notebook
105             60             20 115                          14

P016

PRODUCT
-------
PRODUCT_NAME
```

4. Implement RIGHT JOIN

1. Display all orders with customer information.

```
SQL> SELECT c.customer_id, c.customer_name, o.order_id, o.order_date, o.total_amount FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id;

CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------------------------
ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- --------- ------------
        101 Arun Kumar Raj
111                  10-JAN-25         2500

        102 Priya Sharma
112                  15-FEB-25          307

        103 Rahul Raj
113                  20-MAR-25         4800


CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------------------------
ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- --------- ------------
        104 Divya Devi
114                  05-APR-25          400
```

2. Find orders where customer details are missing.

```
SQL> SELECT c.customer_id, c.customer_name FROM customer c WHERE NOT EXISTS (SELECT 1 FROM orders o JOIN order_details od ON o.order_id = od.order_id WHERE o.customer
_id = c.customer_id);

no rows selected
```

3. Display all payment records with order details.

```
SQL> SELECT p.product_id, p.product_name, od.order_id, od.quantity, od.price FROM product p LEFT JOIN order_details od ON p.product_id = od.product_id;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
ORDER_ID             QUANTITY      PRICE
-------------------- ---------- ---------
P001
Ball Pen
111                         5       1000

P002
Gel Pen
112                        10       2000

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
ORDER_ID             QUANTITY      PRICE
-------------------- ---------- ---------

P003
Pencil
113                        15       3000

P004
Pencil Box

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
ORDER_ID             QUANTITY      PRICE
-------------------- ---------- ---------
114                        20       4000

P005
Spiral Notebook
115                        14       1500

P016

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
ORDER_ID             QUANTITY      PRICE
-------------------- ---------- ---------
Notebook


6 rows selected.
```

5. Retrieve Complete Order Details

Generate a complete order report by combining:

- Customer Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME, OD.QUANTITY, O.ORDER_DATE, O.TOTAL_AMOUNT, PY.PAYMENT_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOME
R_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID LEFT JOIN PAYMENT PY ON O.ORDE
R_ID = PY.ORDER_ID;

CUSTOMER_NAME        PRODUCT_NAME         QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
-------------------- -------------------- -------- --------- ------------ ---------------
Arun Kumar Raj       Ball Pen                    5 10-JAN-25         2500 Success
Priya Sharma         Gel Pen                    10 15-FEB-25          307 Success
Divya Devi           Pencil Box                 20 05-APR-25          400 Success
Rahul Raj            Pencil                     15 20-MAR-25         4800 Pending
```

6. Display Customer Purchase History

1. All products purchased by a customer.

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME, OD.QUANTITY FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.O
RDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID WHERE C.CUSTOMER_ID = 101;

CUSTOMER_NAME        PRODUCT_NAME         QUANTITY
-------------------- -------------------- --------
Arun Kumar Raj       Ball Pen                    5
```

2. Total amount spent by each customer.

```
SQL> SELECT C.CUSTOMER_NAME, SUM(TOTAL_AMOUNT) AS TOTAL_SPENT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,
 C.CUSTOMER_NAME;

CUSTOMER_NAME        TOTAL_SPENT
-------------------- -----------
Rahul Raj                   4800
Priya Sharma                 307
Divya Devi                   400
Arun Kumar Raj              2500
```

3. Number of orders placed by each customer.

```
SQL> SELECT C.CUSTOMER_NAME, COUNT(ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER
_ID, C.CUSTOMER_NAME;

CUSTOMER_NAME        TOTAL_ORDERS
-------------------- ------------
Rahul Raj                       1
Priya Sharma                    1
Divya Devi                      1
Arun Kumar Raj                  1
```

4. Latest purchase details of customers.

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.ORDER_D
ATE = (SELECT MAX(O2.ORDER_DATE) FROM ORDERS O2 WHERE O2.CUSTOMER_ID = C.CUSTOMER_ID);

CUSTOMER_NAME        ORDER_ID             ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- --------- ------------
Arun Kumar Raj       111                  10-JAN-25         2500
Priya Sharma         112                  15-FEB-25          307
Rahul Raj            113                  20-MAR-25         4800
Divya Devi           114                  05-APR-25          400
```

7. Generate Multi-Table Business Reports

Report 1: Customer Order Report

Display:

- Customer name
- Order ID
- Order date
- Order status

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME        ORDER_ID             ORDER_DAT ORDER_STATUS
-------------------- -------------------- --------- --------------------
Arun Kumar Raj       111                  10-JAN-25 Delivered
Priya Sharma         112                  15-FEB-25 Shipped
Rahul Raj            113                  20-MAR-25 Pending
Divya Devi           114                  05-APR-25 Pending
```

Report 2: Sales Report

Display:

- Product name
- Quantity sold
- Total revenue

```
SQL> SELECT P.PRODUCT_NAME, SUM(OD.QUANTITY) AS QUANTITY_SOLD, SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE FROM PRODUCT P JOIN ORDER_DETAILS OD ON
P.PRODUCT_ID = OD.PRODUCT_ID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
Ball Pen                         5          5000
Spiral Notebook                 14         21000
Pencil                          15         45000
Pencil Box                      20         80000
Gel Pen                         10         20000
```

Report 3: Payment Analysis Report

Display:

- Payment mode
- Number of transactions
- Successful payments

```
SQL> SELECT PAYMENT_MODE, COUNT(*) AS NUMBER_OF_TRANSACTIONS, SUM(CASE WHEN PAYMENT_STATUS = 'Success' THEN 1 ELSE 0 END) AS SUCCESSFUL_PAYMENTS FRO
M PAYMENT GROUP BY PAYMENT_MODE;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- --------------------
Net Banking                               1                    0
Card                                      1                    1
Cash                                      1                    1
UPI                                       2                    2
```

Report 4: Customer Purchase Analysis

Find:

- Top purchasing customers

```
SQL> SELECT C.CUSTOMER_NAME, SUM(OD.QUANTITY) AS TOTAL_QUANTITY_PURCHASED FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_
DETAILS OD ON O.ORDER_ID = OD.ORDER_ID GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME ORDER BY TOTAL_QUANTITY_PURCHASED DESC;

CUSTOMER_NAME        TOTAL_QUANTITY_PURCHASED
-------------------- ------------------------
Divya Devi                                 20
Rahul Raj                                  15
Priya Sharma                               10
Arun Kumar Raj                              5
```

- Customers with maximum orders

```
SQL> SELECT C.CUSTOMER_NAME, COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID
, C.CUSTOMER_NAME ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_NAME        TOTAL_ORDERS
-------------------- ------------
Rahul Raj                       1
Arun Kumar Raj                  1
Divya Devi                      1
Priya Sharma                    1
```

- Customers with highest spending

```
SQL> SELECT C.CUSTOMER_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOME
R_ID, C.CUSTOMER_NAME ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME        TOTAL_SPENDING
-------------------- --------------
Rahul Raj                      4800
Arun Kumar Raj                 2500
Divya Devi                      400
Priya Sharma                    307
```